BLEED NO MORE: GENERATIVE INTERFERENCE REDUCTION FOR MUSICAL RECORDINGS

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Abstract—Live multitrack recordings often contain microphone bleed, where unintended sources contaminate close-mic tracks. We cast interference reduction as conditional generation of the target signal from its contaminated observation. Our method, cWGAN-IR, uses a conditional adversarial model with a U-Net generator and a patch-based critic trained on time-frequency magnitude spectrograms. Trained on MUSDB18HQ with simulated bleed and evaluated under both simulated and re-recorded room conditions, the approach yields consistent gains in scale-invariant signal-to-distortion ratio (SI-SDR) and signal-to-interference ratio (SIR) over classical interference reduction baselines for vocal, bass, and drums. The model shows encouraging transfer to Indian classical multitracks (Sanidha) and to live Saraga recordings in a qualitative setting, suggesting robustness to instrument and style mismatch. Taken together, the results suggest that conditional generative modeling is a viable approach for multitrack interference reduction in realistic acoustic conditions.

Index Terms— Bleed Reduction, Music Source Separation, Interference Reduction, Adversarial Learning

1. INTRODUCTION AND RELATED WORK

Live multitrack recordings often contain microphone bleed, where sound from non-target sources leaks into a close-mic channel and reduces the usefulness of stems for editing, mixing, and model training. Unlike music source separation, which aims to reconstruct every source, interference reduction focuses on a single target channel and suppresses cross-bleed while preserving timbre and transients. The recording setup differs from standard array processing: each microphone is placed close to its intended source, yielding high source-to-interference ratio, but microphones are spaced several meters apart across the stage (Figure 1). This wide spacing causes spatial aliasing and breaks key assumptions of array beamforming (which requires small inter-mic spacing relative to wavelength) and of blind source separation methods (which rely on statistical independence across channels) [1–8]. As a result, classical multichannel pipelines degrade in real venues, making an independent channel-wise approach that exploits the close-mic advantage attractive.

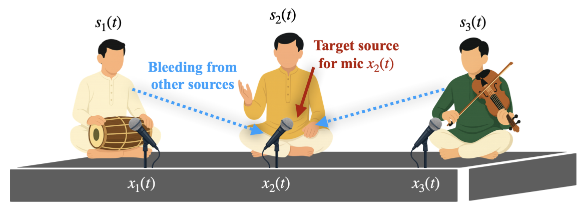

Fig. 1. Illustration of source bleeding in a live Carnatic concert. The three performers, a mridangam player, a vocalist, and a violinist, are recorded using close microphones.

Interference reduction can be viewed as a special case of music source separation, which is well studied. End-to-end architectures such as Hybrid Transformer Demucs and Band Split RNNs achieve strong benchmark scores on MUSDB18HQ [9, 10], but they are optimized to output all instrument stems. This can alter the fidelity of the target track when the actual goal is simply to suppress bleed and preserve the target channel. Generalization to out-of-domain material such as Indian classical music remains limited due to dataset shift and the scarcity of clean, genre-diverse multitracks; live concert corpora like Saraga illustrate the challenge [11].

Interference reduction has therefore been studied as channel-wise post-processing. Early methods used Wiener filtering or NMF-based spectral estimation with generalized Wiener filtering [12, 13]. KAMIR remains a strong baseline but is computationally intensive for large datasets and real-time processing [14], while approximations such as Fast-MIRA trade accuracy for speed [15]. Deep regression models such as CAE and t-UNet learn masks from contaminated inputs, but their reliance on instantaneous mixing and limited high-frequency modeling can leave audible bleed [16]. Spatial filtering and blind separation methods (e.g., MVDR, AuxIVA) work when array geometry, independence, and stationarity hold, yet stage layouts with wide spacing, spatial aliasing, and strong correlations among musical stems violate these assumptions in practice [1, 2].

Generative enhancement brings a complementary toolset. In speech and music, adversarial and diffusion-based en-

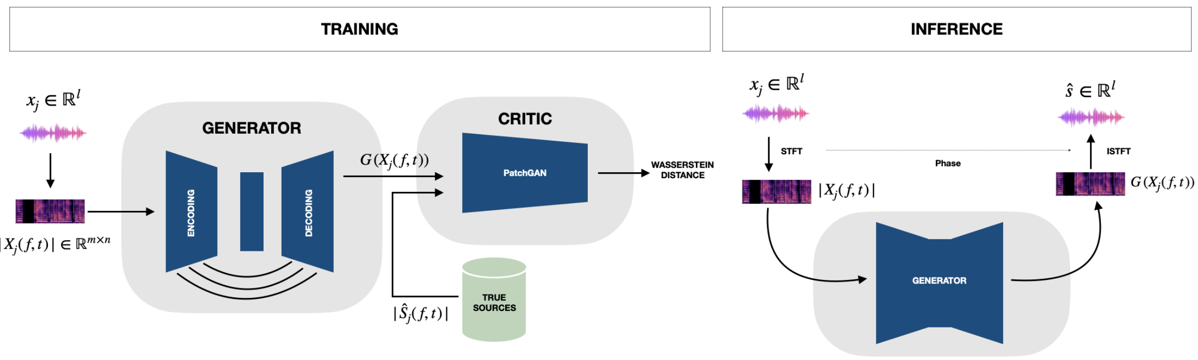

Fig. 2. Overview of the cWGAN-IR interference reduction framework. Left: Training pipeline with U-Net generator, PatchGAN critic (Wasserstein + GP) and L_1 loss. Right: Inference pipeline with generator output, phase reuse, and ISTFT.

hancement improve perceptual realism by matching local spectral texture rather than minimizing only pointwise errors [17–19]. Patch-based critics promote faithful high-frequency structure [20, 21], and Wasserstein objectives with gradient penalty stabilize training [22]. These ideas have not been systematically positioned for multitrack interference reduction with a single target channel and realistic room transfer.

We address interference reduction as conditional generation: Given a contaminated close-mic observation of a target source, the system generates an enhanced version that maintains musical texture while attenuating bleed. Our method, cWGAN-IR, couples a U-Net generator with a patch critic and trains with a Wasserstein objective and an L1 fidelity term on time–frequency magnitudes, with waveform reconstruction at inference. We evaluate on MUSDB18HQ with simulated bleed spanning realistic studio and live acoustics, a re-recorded subset that captures real room transfer, and transfer tests on Indian classical multitracks. Results show consistent gains over classical interference reduction baselines and competitive behavior relative to a strong separation model, with a clear trade-off in signal-to-artifacts ratio (SAR) that reflects mixture-phase reconstruction and sharper spectral contrasts.

Contributions. We cast multitrack interference reduction as conditional generation for a single target channel under convolutive mixing, distinct from source separation; introduce cWGAN-IR, a compact adversarial model that suppresses bleed while preserving target fidelity; and evaluate on MUSDB18HQ with simulated and re-recorded bleed, including ablations on adversarial training and standard metrics (SI-SDR, SIR, SAR, SNR).

2. MATHEMATICAL MODELING

Let $s_i(t)$ denote the ith source for $i = 1, \dots, N$ and let $x_j(t)$ denote the recording at microphone j for $j = 1, \dots, N$. We assume one microphone per source and that source j is dominant at microphone j. Each signal is a convolutive mixture with additive noise,

$$x_j(t) \;=\; \sum_{i=1}^{N} \big(s_i * h_{ij}\big)(t) \;+\; v_j(t), \tag{1}$$

where $h_{ij}(t)$ is the acoustic impulse response from source i to microphone j and $v_j(t)$ is additive noise. Writing the dominant and interference terms explicitly,

$$x_j(t) \;=\; \underbrace{\big(s_j * h_{jj}\big)(t)}_{\text{desired}} \;+\; \underbrace{\sum_{i \neq j} \big(s_i * h_{ij}\big)(t) + v_j(t)}_{\text{interference}}. \tag{2}$$

For analysis we use the short-time Fourier transform (STFT). We model the direct path from source j to microphone j by a slowly varying transfer function $H_{jj}(f)$ and absorb late reverberation and all interferers into $I_j(f,t)$:

$$X_j(f,t) \;=\; S_j(f,t)\, H_{jj}(f) \;+\; I_j(f,t), \tag{3}$$

where X_j and S_j are the STFTs of x_j and s_j. Our objective is to recover the interference-reduced STFT

$$\hat{S}_j(f,t) \;=\; S_j(f,t)\, H_{jj}(f), \tag{4}$$

from the observation $X_j(f,t)$ using a data-driven conditional generator. In practice, the generator operates on magnitude spectrograms: given $|X_j(f,t)|$, it predicts an estimate $G(|X_j|) \approx |S_j(f,t)H_{jj}(f)|$, which is then combined with the mixture phase $\angle X_j(f,t)$ for waveform reconstruction.

3. ARCHITECTURE

We treat interference reduction as conditional generation on time-frequency magnitudes. Let $|X_j(f,t)|$ be the magnitude spectrogram of the contaminated close-mic channel for target j and $|S_j(f,t)H_{jj}(f)|$ the corresponding clean target magnitude. The generator G maps $|X_j|$ to an estimate

$G(|X_j|) \approx |S_j H_{jj}|$ (Figure 2). At inference the waveform is reconstructed as $\tilde{s}_j = \mathrm{iSTFT}(G(|X_j|), \angle X_j)$.

The generator is a U-Net with three downsampling stages (channel sizes 64-128-256), a dilated bottleneck, and skip connections. Each encoder block uses two Conv-BN-ReLU layers with strided downsampling; the bottleneck applies dilations of two and four to enlarge receptive field. The decoder mirrors the encoder with transposed convolutions and skip concatenation to preserve spectral detail, followed by a 1×1 conv output. Direct magnitude regression proved stable in our setting.

The critic D is a PatchGAN-style network [20] that scores overlapping spectrogram regions. The input is the channel-wise concatenation of $|X_j|$ with either $|S_j H_{jj}|$ or $G(|X_j|)$. D comprises strided conv-LeakyReLU-IN layers (except first/last) and outputs local scores averaged across patches.

We adopt the WGAN-GP formulation [22, 23]. The critic loss is

$$\begin{aligned}\mathcal{L}_D = \mathbb{E}[D(G(|X_j|), |X_j|)] - \mathbb{E}[D(|S_j H_{jj}|, |X_j|)] \\ + \lambda_{\mathrm{gp}}\, \mathbb{E}(\|\nabla_{S^\epsilon} D(S^\epsilon, |X_j|)\|_2 - 1)^2 \,,\end{aligned} \tag{5}$$

with $S^\epsilon = \epsilon|S_j H_{jj}| + (1-\epsilon)G(|X_j|)$, $\epsilon \sim \mathcal{U}[0,1]$, $\lambda_{\mathrm{gp}} = 10$. The generator is trained with

$$\mathcal{L}_G = -\,\mathbb{E}[D(G(|X_j|), |X_j|)] + \alpha \|\, |S_j H_{jj}| - G(|X_j|)\,\|_1, \tag{6}$$

where α balances fidelity and realism. Training expectations are over examples and interpolations in $\mathcal{L}_D$.

4. EXPERIMENTAL EVALUATION

4.1. Datasets and Experiments

We evaluate across four settings that span simulated, re-recorded, and live recordings: (i) MUSDB18HQ with simulated bleed (**MUSDB18HQ-S**) [24]; (ii) MUSDB18HQ re-recorded in a room (**MUSDB18HQ-R**); (iii) **Sanidha** multitracks (Carnatic music) [25]; and (iv) **Saraga** live concert recordings (no clean stems) [11]. All simulated interference mixing follows Equation 1.

MUSDB18HQ-S (simulated bleed). We use three stems (vocal, bass, drums). Tracks are split into 10 s segments with 50% overlap. To synthesize realistic interference, we simulate shoebox rooms with `pyroomacoustics` [26]. Room dimensions are sampled from length 5–12 m, width 4–8 m, height 2.7–4 m; RT60 is drawn from 0.2–0.6 s. The target source has a dedicated close mic at 0.2–0.5 m; interfering sources are placed 1–3 m away at random azimuths. Source gains are randomized so that input SIR spans a broad range. After energy-based activity detection (retain segments with ≥ 2 s continuous activity), we obtain 7,560 training pairs.

MUSDB18HQ-R (re-recorded bleed). To reduce the mismatch between simulated and real acoustics, a curated subset of MUSDB18HQ was played back on loudspeakers and re-captured with spatially separated microphones (as shown in Figure 1) in the Listening Technology Lab recording studio. Typical placement used ~ 30 cm source-mic spacing for the target and 2–3 m separation among sources to induce real acoustic mixing bleed. This data is used only for evaluation.

Sanidha (Carnatic multitracks). Sanidha provides studio-quality multitrack recordings with clean stems (vocal, mridangam, violin, ghatam) [25]. We create mixtures using the same 10 s/50% overlap protocol as MUSDB18HQ-S and use 2,300 pairs for training/evaluation to test cross-cultural generalization.

Saraga (live Carnatic concerts). Saraga consists of professionally recorded live performances [11]. Clean references are unavailable; we therefore use Saraga for *qualitative* listening and case studies, not for reference based metrics.

Preprocessing and training. All audio stems in MUSDB18HQ are provided as stereo recordings; we resample them to 22.05 kHz and sum to mono for training. We compute STFTs with a Hann window of 2047 samples and a hop of 512 samples; the odd window length keeps down/upsampling in the U-Net exactly aligned. Magnitude spectra are used as inputs. Unless noted, models are trained for 600 epochs with Adam (lr $= 2 \times 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.999$), batch size 8, and main loss weight $\alpha = 10$.

4.2. Results and Discussion

We compare the proposed method against established interference reduction baselines: KAMIR [14], CAE [16], and t-UNet [16], as well as the source separation model HTDemucs [9]. For context, we also include a reference row comparing clean stems against their corresponding bleed mixtures, which indicates the raw contamination level. Evaluation metrics are SI-SDR, SNR, SIR, and SAR. For magnitude-only systems we reconstruct waveforms using the input phase, while waveform models use their native outputs.

Table 1 reports MUSDB18HQ-S and -R results for vocal, bass, and drums. The proposed cWGAN-IR surpasses classical interference reduction baselines across sources and metrics on both simulated and re-recorded sets, with clear gains in SI-SDR, SNR, and SIR. Relative to HTDemucs, cWGAN-IR is competitive on MUSDB18HQ-S and shows an advantage on MUSDB18HQ-R in SI-SDR, SNR, and SIR, while HTDemucs attains higher SAR, which is consistent with our use of mixture-phase reconstruction and the sharper spectral contrasts induced by magnitude-only generation.

Table 2 evaluates cross-domain transfer to Sanidha (vocal, mridangam, violin, ghatam). HTDemucs shows domain mismatch, especially on vocal, whereas our model, trained primarily on Western stems plus simulated bleed, retains stronger SI-SDR and SIR, indicating better suppression without over smoothing. HTDemucs results for mridangam, violin, and ghatam are not directly comparable because its

Table 1. Results on MUSDB18HQ for simulated (S) and real (R) conditions. Mean across the examples. Higher is better.

	Vocal				**Bass**				**Drums**			
Method	SI-SDR	SNR	SIR	SAR	SI-SDR	SNR	SIR	SAR	SI-SDR	SNR	SIR	SAR
						MUSDB18HQ-S						
Reference	-23.42	-2.69	23.54	24.58	-14.25	-1.59	34.47	19.79	-19.07	-0.48	27.00	17.07
KAMIR	4.53	3.88	6.92	5.47	6.18	5.42	7.00	6.27	3.15	2.94	4.21	3.56
CAE	9.12	8.66	9.77	8.90	7.91	7.36	9.82	7.64	6.88	6.12	9.03	7.45
t-UNet	-22.67	-2.15	24.56	24.91	-13.72	-1.18	34.89	20.16	-18.49	-0.33	27.26	17.42
HTDemucs	**16.36**	**16.45**	37.93	**24.23**	16.87	5.61	40.92	**24.69**	13.09	3.13	25.36	**20.77**
cWGAN-IR	13.09	9.96	**38.64**	17.48	**17.38**	**10.64**	**42.44**	16.59	**13.87**	**10.29**	**29.72**	15.60
						MUSDB18HQ-R						
Reference	-31.97	-2.65	12.46	-1.81	-20.65	-1.96	9.16	-4.80	-32.11	-0.67	7.71	-2.04
KAMIR	1.02	-0.43	2.58	0.84	-0.67	1.21	2.73	1.12	-2.76	0.35	1.97	-0.28
CAE	0.94	1.15	3.78	2.26	1.48	0.33	3.67	1.57	-2.92	0.06	3.29	0.73
t-UNet	-31.22	-2.31	12.69	-1.42	-19.94	-1.62	9.48	-4.39	-31.53	-0.49	7.88	-5.63
HTDemucs	-8.46	0.64	21.89	**11.27**	-6.29	0.67	20.67	**3.36**	-8.58	0.94	18.39	**5.06**
cWGAN-IR	**2.30**	**3.52**	**22.79**	3.62	**2.02**	**2.10**	**22.74**	3.24	**-2.58**	**1.66**	**18.79**	-0.90

Table 2. Results on the Sanidha dataset. Mean across the examples. N/A marks instruments not produced by CAE/HTDemucs.

	Vocal				**Mridangam**				**Violin**				**Ghatam**			
Method	SI-SDR	SNR	SIR	SAR	SI-SDR	SNR	SIR	SAR	SI-SDR	SNR	SIR	SAR	SI-SDR	SNR	SIR	SAR
True vs Bleed	0.29	0.27	0.84	15.71	9.05	9.03	19.38	25.51	5.65	5.60	6.25	19.50	-3.32	-3.56	-1.90	13.54
KAMIR	1.12	0.86	6.42	5.37	2.84	1.71	10.35	9.58	4.89	4.72	7.48	8.96	0.52	1.33	4.56	4.02
CAE	0.68	1.42	9.82	4.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
t-UNet	0.74	0.61	1.36	**15.98**	8.46	8.25	19.92	**25.89**	5.92	5.84	6.73	**19.76**	-3.01	-3.15	-1.42	**13.78**
HTDemucs	-26.13	-2.59	-17.50	9.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
cWGAN-IR	**3.46**	**3.21**	**13.51**	5.87	**9.49**	**9.93**	**26.28**	21.08	**7.71**	**7.83**	**16.88**	10.31	**1.18**	**1.60**	**9.81**	4.70

Table 3. Metrics are mean across tracks and sources.

	MUSDB18HQ-S		**MUSDB18HQ-R**	
Method	**SI-SDR**	**SIR**	**SI-SDR**	**SIR**
ℓ_1 only	12.05	30.51	0.18	20.65
ℓ_1+adv	14.78	36.93	0.58	21.44

fixed output stems (vocals, drums, bass, other) do not map to these instruments.

For Saraga (live concerts without references) we provide qualitative audio examples online.[1] Informal listening suggests clearer target prominence and fewer bleed artifacts than other models, particularly in dense percussion passages; formal listening tests are left to future work.

4.3. Ablation studies

We compare an ℓ_1 only generator against ℓ_1 plus adversarial training under identical data and schedules. Evaluation uses mean SI-SDR and SIR over vocal, bass, and drums on simulated and rerecorded MUSDB bleed. Adversarial training improves SIR and yields a smaller but consistent gain in SI-SDR (Table 3).

[1] Audio examples: `listeningtech.github.io/cGANIR/`

5. CONCLUSION

We presented a generative approach to musical interference reduction using a conditional Wasserstein GAN with gradient penalty. In contrast to classical signal processing or source separation systems that embed mixing or source priors, we treat interference reduction as conditional generation in the time-frequency domain: Given a contaminated spectrogram with a dominant target per channel, the model reconstructs a cleaner target without explicit room or geometry models.

Across MUSDB18HQ with simulated and re-recorded bleed and the Sanidha Carnatic multitracks, the method outperforms interference reduction baselines such as KAMIR, CAE, and t-UNet on SI-SDR, SNR, and SIR. It remains competitive with a strong separation network on matched material and shows better behavior on non-Western instruments. Qualitative examples on Saraga live concerts further suggest improved target prominence and reduced bleed.

Limitations remain. Mixture phase reconstruction can reduce SAR relative to methods that estimate phase. Future work will incorporate phase-aware or complex time-frequency modeling, exploit multi-microphone spatial cues and learned room priors, and pursue efficient streaming inference. We view conditional generation as a compact and genre-agnostic foundation for interference reduction in varied acoustic environments.

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